

We do not build aircraft. We build what turns one into a working system.

Between a flying platform and a decision someone has to make on the ground there is a layer of software: mission control, autonomy, perception, and reporting. That layer is our product. **Whatever you fly, we make it operate, see, and report** — under your name, on your platforms, with your data staying yours.

A software house, not a systems integrator with a slide deck.

BBA Technology builds and ships its own software — twenty-eight products across unmanned systems, hospitality, and sports, written in-house. Every module in this file carries a label: *Deployed* is running for a customer right now. *R&D programme* is a funded line in our own laboratory, available on a project basis and ahead of where the market is. Both are ours, and we say which is which.

Command and control

Mission planning, live telemetry, multi-aircraft and ground-robot control, swarm operation, and tasking in plain language. Page 2.

Perception and analytics

Detection, counting, density, change between flights, digital twins, and processing that runs on the aircraft itself. Page 3.

Reporting and compliance

Dated reports an institution can file, role-based access, an audit log that cannot be altered, and on-premises deployment where required.

Delivered under your name

White-label on your platforms and in your interface, as a per-site licence with a monthly fee or as a fixed-scope project. Page 4.

WHERE THE WORK STANDS TODAY

UAE municipality

coastal monitoring — fifteen analyses from one flight, on-premises

303,000 m²

security and operations platform, major retail development, Dubai

Municipal

excavation and infrastructure monitoring, metropolitan municipality

Energy

aerial inspection for a national petroleum corporation

PART ONE — OPERATING THE PLATFORM

One interface for everything that moves: air, ground, and the sensors that are already fixed to the site.

Ground control station

DEPLOYED

Mission planning, live telemetry, and flight control for several aircraft at once, on one screen. Areas are drawn on the map, the flight plan is generated from the area rather than from waypoints typed by hand, and the same mission can be repeated on a schedule so that two flights over the same ground stay comparable.

Integrates with open autopilot telemetry and major commercial flight platforms. **Runs** on the operator's own machine or server; no dependency on a vendor cloud.

Ground robots on the same platform

DEPLOYED

Ground vehicles and service robots are commanded from the same interface as the aircraft, with the same mission and reporting model. An operator who has learned the air side does not learn a second system for the ground side.

Deployed in a commercial logistics environment with autonomous ground units.

Command by message

DEPLOYED

A mission can be tasked, and its result received, in plain language from a phone — no operator console, no training. The person who needs the answer asks for it directly instead of asking the team that operates the system.

Why it matters: most unmanned programmes fail on adoption, not on technology. The people who need the output are rarely the people trained to fly.

Swarm operation

R&D PROGRAMME

Several aircraft sharing one mission: an area is divided between them, coverage is tracked jointly, and one operator supervises the group rather than each aircraft. Built and flying in our own laboratory, and the natural extension of the multi-aircraft control already deployed.

Available as a project delivery on a defined site.

Flight without satellite positioning

R&D PROGRAMME

Visual and inertial navigation for indoor, tunnel, and denied environments, so that position is held from what the aircraft sees rather than from what it receives. A continuously funded research line — one of the hardest problems in the field, and one we have chosen to own rather than buy.

Available as a joint development with a partner who has the platform and the operating environment.

PART TWO — TURNING FLIGHT INTO AN ANSWER

A recorded flight is not information. These modules are what convert it into a number an institution can file, compare, and act on.

Detection, counting, and density

DEPLOYED

People, vehicles, vessels, and objects detected in aerial imagery, counted, and turned into a density map. The output is a figure with a location and a time, not a video for someone to watch.

Ten object classes in high-altitude imagery. **Runs on ordinary CPU hardware** — deployable in the field without a GPU workstation, which matters more than raw speed when the site has no data centre.

Change between two flights

DEPLOYED

The same area flown twice, compared automatically: what was added, what was removed, how far a line moved. This is what makes monitoring a measurement rather than an opinion, and it is the basis of the municipal and coastal work below.

Delivered for excavation and infrastructure monitoring, where the dated difference between two flights is the deliverable.

Digital twin of the site

DEPLOYED

Flight footage rebuilt into a three-dimensional model of the terrain or facility, so that distance, area, volume, and surface change can be measured on the model instead of estimated from pictures.

AI that runs on the aircraft

DEPLOYED

Processing on the onboard computer rather than on a server, so the system keeps producing results when the link to the ground is degraded or lost, and so imagery does not have to leave the platform to be useful.

Reporting, automatically

DEPLOYED

Every flight produces a dated report — imagery, counts, differences, exceptions — in the form an institution can file and audit. Role-based access, an audit log that cannot be altered, and encrypted storage are part of the platform, not an afterthought.

Deployed on-premises where required, with the customer's own database and reporting tools, and no data leaving their network.

Security and operations platform

DEPLOYED

Aerial coverage, fixed cameras, alarms, and patrol routes brought into a single operations picture for a site, a campus, or a coastline — the layer above the individual sensors, where an operator sees the place rather than the equipment.

Delivered for institutions, not demonstrated at fairs.

Municipal authority, United Arab Emirates

Coastal monitoring from the air: crowd counting, zone and vehicle violations, fire and waste detection, water anomalies, asset condition, shoreline and erosion — fifteen analyses from a single flight, each against a numerical acceptance threshold agreed in the contract, running entirely on the customer's own servers.

AWARDED · IN DELIVERY

Retail development, Dubai

Security and operations platform for one of the largest retail developments in the world — 303,000 m² of gross leasable area. Cameras, alarms, patrol routes and aerial coverage in one operations picture.

IN BUILD

Metropolitan municipality, Türkiye

Excavation and infrastructure-work monitoring: repeated flights over the same sites, change measured between them, dated reports filed.

DELIVERED

National petroleum corporation

Aerial inspection and imaging over energy infrastructure.

DELIVERED

How we deliver

Your aircraft stay

We integrate with the platforms you already fly. We do not sell airframes and do not make them — that is a deliberate choice.

Under your name

The software can be delivered as your product, in your interface, for your customers. We stay behind it and do not appear to the end user.

On your servers

On-premises and offline operation where the customer requires it, with their own database, their own access control, and no data leaving the network.

Licence or project

Either a per-site licence with a monthly fee, or a fixed-scope project delivery. Both include the source of what is built for you.

One flight recording is enough to start.

Send us a video from an operation you already run. We will process it — detection, counting, change, mapping — and return the output together with the report it generates, so your technical team can judge the system on your own area rather than on ours. That is how every engagement in this file began.

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